Comp. Phys. Lab for QM
Winter Quarter 2024
Physics 115L

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Physics 317

Lecture: Thursday 3:10-4:00 PM in Olson Hall 250
(West of Shields Library)

Textbooks: Intro. to Quantum Mechanics (3rd Ed.) by Griffiths and Schroeder
(Recommended for reference, or any other Intro. QM text.)
Lecture notes on Scientific Python (link on course site)

Course TAs:	TBD	(TBD@ucdavis.edu)
Office Hours:	Mulhearn:	Thursdays 1:00-2:00 PM PHY 317
	Wells:	Tuesdays and Wednesdays (Details on course website)

Exams/Quizzes: None planned, but see caveats in the plagiarmism section.
Final Project: Due March, 21 2025 1:00 PM

Course Description:
Applications of computational physics to problems from Classical and Quantum Mechanics. You will bring the Schrödinger Equation to life! Application of numerical techniques, and comparison to analytic solutions (where they exist). Visualization of the wave function and its time evolution. Applications of the Discrete Fourier transform.

Plagiarism and Academic Conduct:
You must adhere to the UC Davis Code of Academic Conduct, available online at ossja website. In particular, you may not take credit for any work not created by you. **You may not direct search engines, generative AI tools, homework help websites, or other external resources toward the specific problems that are assigned to you.** If you do, then you must disclose it, and you will receive a zero for the entire assignment. If you do not disclose it, and are caught, then you will fail the entire course.

But how will I know? It is not easy, but we are learning, and I am working with your TAs on this issue. If I suspect there is widespread non-compliance with this policy, then I will implement prove-you-wrote it quizzes. If I or the TAs notice anything suspicious, we will schedule oral exams with specific students on short notice. I also reserve the right to have a final exam if I suspect widespread abuse. If we find evidence of academic misconduct, we will refer the matter to OSSJA. Students found to have committed academic misconduct will fail the course.

You are allowed to collaborate with other students in the course, including discussing specific problems in the homework. But the solution that you provide must be your own: each and every line. In general, you should be able to explain why each line in your code is there. If you do not understand something in your solution because you relied on trial and error, then you should state that in the comments, including explaining all of the things that you tried.

Why do I care? You are under intense time pressure, and the university is in one of its roles a gatekeeper to your future career. It seems like thrashing about is such a waste of time, when you can find the answer online and take a more direct route to providing the deliverable I am looking for. Here's the problem: the thrashing about **is** the deliverable I am looking for. Why? Because I want you to learn how to do difficult things, and the only way, so far, that humans have found how to learn difficult things is by thrashing about. There aren't any shortcuts. So don't rely on external resources to complete your assignments. Put in the work, and a bright future awaits you!

Course Philosophy:

Computational physics is essential to nearly every career related to physics. Unfortunately, the educational pipeline, since grade school, does not adequately prepare students with programming skills, and so a central feature of computational physics courses is a wide distribution of student programming abilities. There are students that are better programmers than most physics professors, students that have little practice outside of the required computational courses, and everything in between.

Fortunately, in this course we are concerned with **computational physics** and not just **computer programming**. You do not need to be an expert computer programmer to succeed in this course. You do need to gain enough programming competency to handle the fascinating physics problems we will tackle. **The primary objective of this course is to make you confident in using programming independently as an essential and effective tool for solving physics problems.**

This is a one unit course, and I will do my best to keep the workload appropriate. However, given the wide variation in student programming abilities, inevitably some students will find the homework more time-consuming than others. If you do find yourself working longer on the assignments, please understand that this is time well-spent: gaining more proficiency in computing is almost certainly one of the best possible uses of your time at this stage in your career. There are a vanishing small number of physics-related career options that do not require programming ability.

Homework:
There will be up to five home-work assignments which will feature problems in computational physics.

Final Project:
In lieu of a final exam, you will design an independent creative problem. While your particular problem may have already been solved online, your solution should vary substantially from anything available online.

Grades:
Final grades will be approximately 80% Homework and 20% Final Project. You will be allowed to submit one homework assignment late, but you must complete every assignment to complete the course.